

Report
Of
**ODOO ERP CLIENT PROVISIONING AND
PRODUCTION CI/CD AUTOMATION**

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.
In partial fulfilment of the requirement of degree of
**B.Tech CSE (Specialization:
Cloud Computing)**



By
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CANDIDATE'S DECLARATION

I, Kumar Nirupam, hereby declare that the internship report titled “Odoo ERP Client Provisioning and Production CI/CD Automation” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature

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CHAPTER 3: INTERNSHIP CERTIFICATE

3.1 Official SystemaOps Certificate of Recognition

This is to certify that Kumar Nirupam (Roll No: CS-23411229, Employee ID: 041392381408), a student of B.Tech Computer Science & Engineering (Specialization: Cloud Computing, Section: 4CSE8) at IILM University, Greater Noida, has successfully completed a 3-Month Summer Internship as a Cloud & DevOps Engineering Intern at SystemaOps. The certificate of recognition was awarded on 26-08-2026, signed by Rohit Thumu (Authorized Signatory) and Jhansi (HR Department).



CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

The primary project of my internship was the **automated provisioning of customer-specific Odoo ERP environments using CI/CD pipelines**.

Odoo is used by different businesses for managing their operational requirements through configurable business modules. In a multi-client environment, manually creating and configuring a separate Odoo instance for every customer can become repetitive, time-consuming, and difficult to maintain consistently.

The objective of the provisioning system was to make the onboarding process more structured and automated.

The system connects a client-facing frontend with a CI/CD pipeline. A customer provides their company information and selects the required Odoo modules. This information is then passed into the pipeline, which validates the requirements, determines the required modules from the parent Odoo application, and provisions a client-specific Docker environment.

The resulting environment contains the required Odoo application, database, and reverse-proxy configuration.

The project was particularly important because it connected application onboarding, GitHub Actions CI/CD, Docker containerization, Odoo module management, database isolation, and production deployment into a single workflow.

4.2 Organization Profile

SystemaOps (systemaops.com) is a premier Cloud Infrastructure and DevOps engineering consultancy specializing in cloud-native architectural transformations, container orchestration, Site Reliability Engineering (SRE), and DevSecOps automation. SystemaOps empowers organizations to modernize legacy software delivery into automated, audit-ready, scalable multi-cloud environments.

Operating as a DevOps Engineering Intern at SystemaOps provided immersive, real-world industry experience in maintaining multi-AZ cloud infrastructure, configuring infrastructure state files, building enterprise CI/CD workflows, and diagnosing production container bottlenecks under senior mentorship.

4.3 Problem Statement

Conventional manual deployment paradigms fail to deliver the speed, reliability, and security required by modern multi-tenant enterprise applications. In traditional Odoo ERP customer onboarding, administrators are forced to execute repetitive manual setups ranging from module selection and database provisioning to port allocation, reverse proxy routing, and credential distribution. This manual approach severely increases operational overhead, slows customer onboarding, and leads to frequent configuration inconsistencies across environments.

These deployment delays are further compounded by underlying bare-metal infrastructure challenges. Managing a high-density production environment involves handling active port collisions across competing services, isolating database instances, resolving file permission and repository authentication barriers, and fixing container build failures. Crucially, executing these infrastructure updates without causing downtime for existing production applications presents a major operational obstacle. To eliminate these failure points, the project establishes a fully automated, CI/CD-driven provisioning workflow and container orchestration framework designed to standardize application delivery and guarantee continuous service availability.

4.4 Project Objectives

The primary objective of this project was to transform manual ERP deployments into an automated, multi-tenant cloud delivery platform through the following key targets:

- **Automated Client Provisioning:** Replace manual server setup steps with a streamlined, repeatable provisioning engine.
- **Frontend-Driven CI/CD Workflows:** Connect client module selection inputs directly to GitHub Actions workflow triggers.

- **Centralized Module Sourcing:** Utilize the Parent Odoo Application as a single source of truth for dynamic module selection.
- **Container-Level Tenant Isolation:** Deploy independent Docker containers to keep client application environments completely isolated.
- **Dedicated Database & Reverse Proxy Routing:** Assign separate PostgreSQL database instances and configure Nginx reverse-proxy endpoints for each tenant.
- **Deployment Consistency & Reliability:** Standardize rollout processes to reduce configuration drift and human error.
- **Production Operations & Backup Management:** Maintain high availability for active client instances through automated daily backups, port conflict management, and continuous container health monitoring

4.5 Scope of the Project

The project scope encompassed the complete lifecycle of a multi-tenant Odoo deployment pipeline, bridging initial client requirement collection with continuous production operations. Core responsibilities included tailoring the parent Odoo model, integrating GitHub repositories, and configuring automated CI/CD pipelines for containerized application delivery. The architecture leveraged Docker-based containerization, dedicated PostgreSQL database isolation per tenant, Nginx reverse-proxy traffic management, and dynamic port allocation to ensure system security and operational efficiency. Following successful deployment verification and user access provisioning, operational workflows included maintaining daily automated backups, resolving container bottlenecks, and supporting production operations across secondary applications—gaining extensive hands-on experience with Linux administration, GitHub Actions, and cloud infrastructure.

4.6 Technologies and Tools Used

Table 3.1 Technologies and Tools Used

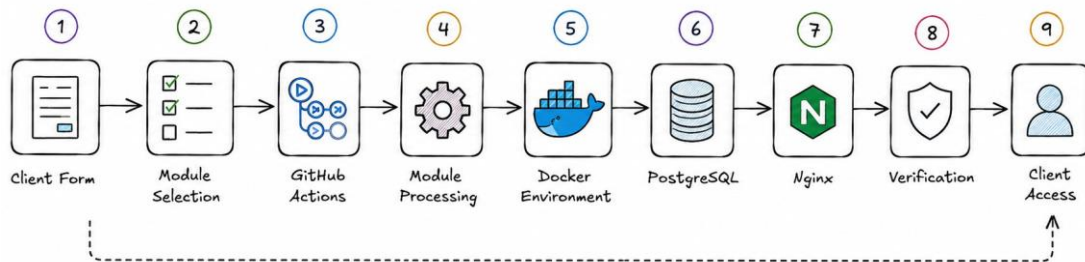
Category	Technology / Tool	Purpose
Operating System	Linux / Debian	Production server environment
Containerization	Docker, Docker Compose	Application isolation and deployment
CI/CD	GitHub Actions	Automated deployment and provisioning
ERP	Odoo 18	Customer ERP application
Database	PostgreSQL	Odoo databases
Reverse Proxy	Nginx	Routing and access
Version Control	Git, GitHub	Source and deployment management
Automation	Shell, Cron	Backups and maintenance

4.7 System Architecture

The Odoo provisioning architecture consists of several connected layers. The client-facing frontend collects company information and required Odoo modules. The submission triggers the GitHub Action CI/CD workflow, which validates the requirements and identifies the required modules from the Parent Odoo Application.

The pipeline then provisions a customer-specific Docker environment. The Odoo application and PostgreSQL database are isolated for the client, while Nginx provides reverse-proxy access. After deployment, the instance is verified and the required access information can be provided to the client.

Figure 1.1 High-Level Odoo Client Provisioning Architecture



4.8 Methodology

The project followed a practical DevOps-based methodology focused on automating the Odoo client provisioning process and maintaining reliable production deployments. The work was carried out in the following stages:

Phase 1: Requirement Analysis and Workflow Understanding

The existing Odoo onboarding process was studied to understand how customer information, module selection, application configuration, database setup, and deployment were connected. The Parent Odoo Application was treated as the central source for the available Odoo modules. The objective was to convert the repetitive manual onboarding steps into a standardized automated workflow.

Phase 2: Client Requirement and Module Selection

A client-facing frontend form was used to collect the company information and Odoo modules required by the customer. The selected modules were passed to the deployment workflow so that the required configuration could be prepared automatically for the specific client.

Phase 3: GitHub Actions CI/CD Automation

GitHub Actions was used as the CI/CD automation layer. The workflow was configured to receive the client deployment requirements and execute the required provisioning steps automatically. This reduced dependency on manually running deployment commands on the production server and provided a repeatable deployment process.

Phase 4: Docker-Based Environment Provisioning

The required Odoo environment was deployed using Docker and Docker Compose. Each client environment was kept isolated through separate Odoo application containers and PostgreSQL database configurations. This approach allowed multiple Odoo environments to operate on the same production server without mixing their application and database data.

Phase 5: Database and Reverse Proxy Configuration

PostgreSQL was configured as the database layer for the Odoo environments. Separate database configurations were maintained for individual client instances. Nginx was used as the reverse proxy to provide controlled access to the deployed Odoo services. Port assignments were checked and adjusted whenever required to prevent conflicts with existing production applications.

Phase 6: Deployment Verification and Troubleshooting

After deployment, container status, application availability, database connectivity, ports, and reverse-proxy configuration were verified. Production issues such as Docker permission problems, port conflicts, repository access problems, database startup issues, and container configuration errors were diagnosed using server commands, Docker logs, configuration files, and service status information.

Phase 7: Production Maintenance and Backup

The deployed Odoo environments were continuously maintained as production services. An automated backup script was created to archive Odoo data daily, retain backups for 30 days, and record backup activity in a log. The backup process was manually tested to verify successful archive creation.

Phase 8: Continuous Improvement

The deployment process was improved based on issues encountered during production operations. Configuration changes, port allocation, Docker resource cleanup, authentication fixes, and deployment workflow improvements were applied to make the infrastructure more reliable and easier to maintain.

4.9 Expected Outcomes

The project was expected to provide the following outcomes:

1. **Automated Odoo Client Provisioning**
Reduce the amount of manual work required to create and configure a new customer-specific Odoo environment.
2. **Repeatable CI/CD Deployment**
Establish a GitHub Actions-based workflow that can execute deployment operations consistently instead of relying on repeated manual server commands.
3. **Isolated Client Environments**
Maintain separate Docker-based Odoo environments and PostgreSQL databases for different clients, reducing the possibility of configuration or data conflicts.
4. **Faster Customer Onboarding**
Reduce the time required to move from customer module selection to a running Odoo environment by automating repetitive provisioning steps.
5. **Consistent Configuration**
Standardize Odoo, PostgreSQL, Docker, Nginx, and port configuration across different client deployments.
6. **Improved Production Reliability**
Support stable operation of multiple Odoo instances through container health checks, systematic troubleshooting, port management, and production maintenance.
7. **Automated Data Backup**
Maintain scheduled backups of Odoo production data with automatic retention and logging to support recovery in case of data loss or service problems.
8. **Reduced Deployment Errors**
Minimize configuration mistakes caused by manually repeating the same deployment procedure for multiple customers.

The internship work resulted in practical production exposure beyond the primary provisioning project. During the internship, six separate Odoo 18 client instances were maintained in Docker, with isolated PostgreSQL databases and Nginx reverse-proxy configurations.

4.10 Certificate of Completion & Communication Proof

A copy of the official Certificate of Recognition, issued by SystemaOps on successful completion of the 3-month "Cloud & DevOps Engineering Internship" (awarded on 26-08-2026), is attached in Chapter 3 of this report. The certificate bears Employee ID 041392381408 and is issued to Kumar Nirupam (Roll No: CS-23411229), authorized and signed by Rohit Thumu (Authorized Signatory) and Jhansi (HR Department).

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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